Volume 5 — Supporting Documents

Principal Investigator Biosketch & Facilities/Equipment Description

Army SBIR Topic ARM26TX04-NV001 | Atelier Logic Group LLC (UEI: G9WFN3AL69J3)

1. Principal Investigator Biosketch: Ishmael Ibrahim Amin

1.1 Contact and Role Information

Principal Investigator: Ishmael Ibrahim Amin **Role:** Principal Investigator & Sole Inventor
Citizenship: United States Citizen
Organization: Atelier Logic Group LLC dba ReluLayer™ **Physical Facility:** 32 Normandy Dr., Kenner, LA 70065
Mailing Address: Atelier Logic Group LLC, Ishmael Amin, 201 Saint Charles Ave., Ste 114 PMB 603, New Orleans, LA 70170
UEI: G9WFN3AL69J3 **CAGE Code:** 20JX8 **Primary NAICS:** 541715 **Primary PSC:** AC11
Official Email: proposals@atelierlogicgroup.com

1.2 Corporate Identifiers

Entity Legal Name: Atelier Logic Group LLC
DBA / Brand: ReluLayer™
UEI: G9WFN3AL69J3
CAGE Code: 20JX8
Primary NAICS: 541715 — Research and Development in Physical, Engineering, and Life Sciences
Primary PSC: AC11 — Defense R&D: Military Sciences / Advanced Development
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Physical Address: 32 Normandy Dr, Kenner, LA 70065-2031 USA
Mailing Address: 201 Saint Charles Ave, Ste 114 PMB 603, New Orleans, LA 70170 US
Sole Inventor: Ishmael Ibrahim Amin

1.3 Education

- **M.S., Data Science in Artificial Intelligence**, Northwestern University, Evanston, IL.
- **B.S., Statistical Science (Actuarial Science)**, University of California, Santa Barbara.

1.4 Professional Experience and Technical Qualifications

Mr. Amin brings 30+ years of enterprise AI, data science, optimization, and large-scale systems integration experience to the proposed effort. His career spans strategic leadership, hands-on engineering, and applied research across defense-relevant and commercial sectors, including aviation, financial services, transit, and federal systems.

Leadership Highlights:

- Former Principal Data Science Strategist and Engineering Leader at DXC Technology, where he led the global AI Guild of 10,000+ members and directly supervised 40+ analytics and data-science personnel across multiple delivery centers.
- Architected custom Mixed-Integer Linear Programming (MILP) gate-assignment optimizers for commercial aviation partners, yielding estimated potential cost savings of $20M–$40M through improved resource utilization and operational scheduling.

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- Led SAS-to-PySpark platform modernizations for tier-1 financial and transit organizations, designing streaming analytics pipelines, real-time event-processing architectures, and cloud-native data platforms.

1.5 Relevance to ARM26TX04-NV001

Mr. Amin's background in optimization, streaming analytics, embedded systems, and secure AI delivery directly supports the proposed work for advanced digital-twin and edge-AI capabilities. His experience bridging high-level machine-learning strategy with low-level embedded implementation ensures the technical approach will be both rigorous and executable within constrained operational environments.

1.6 Intellectual Property Estate

Mr. Amin is the **Sole Inventor** of six Provisional Patent Applications (PPAs I–VI), culminating in **Patent Pending (USPTO Provisional Patent Application VI, Serial No. 64/124,094, Confirmation No. 2105, Priority Date August 1, 2026, Attorney Docket RELU-001-PPA6)**, held by Atelier Logic Group LLC dba ReluLayer™. The PPA VI specification incorporates by reference in their entireties all prior provisional filings (PPA I–PPA V). This portfolio reflects sustained invention in deterministic embedded safety systems, fixed-point projection engines, and hardware-in-the-loop validation frameworks relevant to the proposed SBIR effort.

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2. Facilities and Equipment Description

2.1 Facility Overview

The proposed effort will be executed at the physical engineering facility of Atelier Logic Group LLC dba ReluLayer™, located at **32 Normandy Dr., Kenner, LA 70065**. The on-site work utilizes the **Pacific AI Lab** internal high-performance engineering workstation and testbed environment. The combined facility is a secure, isolated R&D environment dedicated to embedded AI, digital-twin prototyping, and low-level hardware/software co-design. It provides controlled access, conditioned workspace, and a zero-egress network segment for hardware-in-the-loop testing.

2.2 Embedded Hardware Rigs

Atelier Logic Group LLC dba ReluLayer™ maintains multiple ARM Cortex-M7 evaluation platforms at the 32 Normandy Dr. facility, operated within the Pacific AI Lab workstation environment, for prototyping, profiling, and verification of embedded AI and control routines:

- **STM32H743BIT6** evaluation boards (ARM Cortex-M7 @ 480 MHz, 2 MB Flash, 1 MB RAM) for real-time deterministic control and low-latency inference experiments.
- **Pixhawk 6X** flight-controller platforms, used as reference hardware for embedded autonomy, sensor fusion, and safety-critical software workflows.
- Supporting power, interconnect, and breakout peripherals for GPIO, SPI, I2C, UART, and ADC test points.

2.3 Debugging and Emulation Equipment

The lab is equipped with professional-grade debugging and signal-analysis tools:

- **SEGGER J-Link Ultra+** high-speed debug probes for ARM Cortex-M trace and flash programming.
- **ST-Link V3** probes for STM32 on-chip debugging, programming, and virtual COM-port telemetry.
- **USB-Saleae 16-channel** high-speed logic analyzers for protocol capture, timing verification, and edge-case diagnosis.

2.4 Development Environment and Toolchain

- **Software Stack:** ‘#![no_std]‘ Rust toolchain with embedded-hal, cortex-m, and real-time interrupt-driven frameworks; Python/PySpark for data science and offline model training; Tectonic/LaTeX build pipeline for documentation generation.
- **Compute:** High-core-count workstations with CUDA-capable GPUs for model training, simulation, and digital-twin validation; isolated storage for SBIR-controlled technical data.
- **Security:** Isolated zero-egress hardware test rig; version-controlled source under encrypted repositories; access-controlled lab environment.

2.5 Availability

All listed facilities, hardware, and equipment are owned and operated by Atelier Logic Group LLC dba ReluLayer™ and are available for the duration of the proposed SBIR effort without additional lease or rental costs. No third-party facility access is required to execute the proposed technical work.

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3. Teaming and Commercialization Strategy

Atelier Logic Group LLC dba ReluLayer™ will execute the proposed effort through a three-tier team architecture that balances technical leadership, embedded firmware delivery, and dual-use commercialization strategy.

3.1 Three-Tier Execution Architecture

- **Principal Investigator & Systems Architect (Ishmael Ibrahim Amin, M.S.)**: Architecture, PPA VI IP governance, safety manifold formulation, and commercial strategy. Mr. Amin brings 30+ years of enterprise AI, optimization, and large-scale systems integration, including leadership of DXC Technology's 10,000-member AI Guild and estimated $20M–$40M commercial aviation savings from MILP gate-assignment optimization.
- **Senior Embedded / Firmware Engineer (Key Onboarding Role)**: `#![no_std]` Rust kernel, C ABI FFI bridge, and STM32H743 target integration. Outreach with candidate defense embedded software contractors is underway for immediate onboarding upon award.
- **Dual-Use Commercialization Lead (Business Development Advisor)**: Market exploration across FAA Part 107/135 commercial cargo drone operators and Phase II licensing.

3.2 Cross-Industry Transition Alignment

ReluLayer's abstract mathematical framing, governed by PPA VI Section 1.6 (Serial No. 64/124,094), allows the same 326-byte kernel to transition from Army Group 1/2 UAS to commercial logistics drones and autonomous industrial robotics without modification. The safety manifold $\mathcal{S}_v = \{\mathbf{v} \in \mathbb{R}^3 \mid A\mathbf{v} \leq \mathbf{b}\}$ is expressed purely in terms of numerical state vectors and polyhedral constraints, preserving Section 101 eligibility and enabling multi-domain reuse.

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